Extreme Value Theory: Statistical Analysis of Rare Events

Statistical Research Laboratory

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Abstract

This report presents a comprehensive analysis of extreme value theory (EVT) and its applications to rare event prediction. We examine the Fisher-Tippett-Gnedenko theorem, which establishes the generalized extreme value (GEV) distribution as the limiting distribution of block maxima, and the Peaks Over Threshold (POT) method using the generalized Pareto distribution (GPD). Computational analysis demonstrates parameter estimation via maximum likelihood, return level calculation with confidence intervals, threshold selection using mean excess plots, and applications to flood frequency analysis and financial risk assessment.

1 Introduction

Extreme value theory provides the statistical foundation for modeling rare events that lie in the tails of probability distributions. While classical statistics focuses on central tendencies, EVT addresses the behavior of maxima and minima, making it essential for risk assessment in hydrology, finance, insurance, climate science, and engineering.

Definition 1.1 (Extreme Value) *An extreme value is a statistical outlier representing the maximum or minimum of a sample. EVT characterizes the asymptotic distribution of these extremes as sample size approaches infinity.*

2 Theoretical Framework

2.1 The Fisher-Tippett-Gnedenko Theorem

Theorem 2.1 (Fisher-Tippett-Gnedenko, 1928-1943) *Let $X_1, X_2, \ldots, X_n$ be independent, identically distributed random variables with cumulative distribution function F. If there exist sequences of constants $a_n > 0$ and b_n such that*

$$\lim_{n\to\infty} P\left(\frac{\max(X_1, \ldots, X_n) - b_n}{a_n} \leq x\right) = G(x) \tag{1}$$

for a non-degenerate distribution G, then G must be one of three types:

- ***Gumbel*** *(Type I):* $G(x) = \exp\{-\exp[-(x-\mu)/\sigma]\}$, $x \in \mathbb{R}$
- ***Fréchet*** *(Type II):* $G(x) = \exp\{-(x/\sigma)^{-\alpha}\}$, $x > 0$
- ***Weibull*** *(Type III):* $G(x) = \exp\{-[-(x-\mu)/\sigma]^{\alpha}\}$, $x < \mu$

2.2 Generalized Extreme Value Distribution

These three types can be unified into a single parametric family:

Definition 2.1 (GEV Distribution) *The generalized extreme value distribution has cumulative distribution function:*

$$G(x;\mu,\sigma,\xi) = \exp\left\{-\left[1+\xi\left(\frac{x-\mu}{\sigma}\right)\right]^{-1/\xi}\right\} \tag{2}$$

where $\mu \in \mathbb{R}$ is the location parameter, $\sigma > 0$ is the scale parameter, and $\xi \in \mathbb{R}$ is the shape parameter, defined on $\{x : 1+\xi(x-\mu)/\sigma > 0\}$.

Remark 2.1 (Shape Parameter Interpretation) *The shape parameter ξ determines the tail behavior:*

- $\xi = 0$*: Gumbel distribution (light tail, exponentially decaying)*
- $\xi > 0$*: Fréchet distribution (heavy tail, power-law decay)*
- $\xi < 0$*: Weibull distribution (bounded upper tail)*

The limit $\xi \to 0$ yields $G(x) = \exp\{-\exp[-(x-\mu)/\sigma]\}$.

2.3 Peaks Over Threshold Method

Theorem 2.2 (Pickands-Balkema-de Haan, 1975) *For a high threshold u, the distribution of excesses $Y = X - u$ given $X > u$ converges to the generalized Pareto distribution:*

$$H(y;\sigma_u,\xi) = 1 - \left(1+\xi\frac{y}{\sigma_u}\right)^{-1/\xi} \tag{3}$$

where $y \geq 0$ for $\xi \geq 0$ and $0 \leq y \leq -\sigma_u/\xi$ for $\xi < 0$.

Remark 2.2 (Threshold Selection) *Choosing an appropriate threshold u involves a bias-variance tradeoff:*

- ***Too low****: Asymptotic theory invalid, biased estimates*
- ***Too high****: Few exceedances, high variance in estimates*

The mean excess plot aids threshold selection by displaying empirical mean excess $e(u) = E[X-u|X>u]$ versus u.

2.4 Return Levels

Definition 2.2 (Return Level) *The T-year return level x_T is the quantile exceeded on average once every T years:*

$$x_T = \begin{cases} \mu - \frac{\sigma}{\xi}\left[1 - (-\log(1-1/T))^{-\xi}\right] & \xi \neq 0 \\ \mu - \sigma\log(-\log(1-1/T)) & \xi = 0 \end{cases} \tag{4}$$

For the POT method with n_y observations per year and threshold exceedance rate ζ_u:

$$x_T = u + \frac{\sigma_u}{\xi}\left[(Tn_y\zeta_u)^{\xi} - 1\right] \tag{5}$$

3 Computational Analysis

3.1 Simulated Flood Data

4 Results

4.1 Parameter Estimates

4.2 Return Level Analysis

4.3 Method Comparison

5 Discussion

5.1 Interpretation of Results

Example 5.1 (Tail Behavior Classification) *The estimated shape parameter ξ = ?? from the GEV fit indicates a ?? distribution with ??. This suggests that extreme floods have a power-law tail decay, implying significant probability mass in the far tail and potential for catastrophic events.*

Remark 5.1 (Return Level Confidence Intervals) *The 100-year return level is estimated as ?? m^3/s with 95% confidence interval [??, ??] m^3/s. The wide confidence interval reflects parameter uncertainty and underscores the challenges of extrapolation beyond the observed data range.*

5.2 Model Selection: Block Maxima vs. POT

Remark 5.2 (Data Efficiency) *The POT approach utilizes ?? threshold exceedances from ?? daily observations, providing substantially more extreme value information than the ?? annual maxima used in the block maxima approach. This increased sample size generally yields more efficient parameter estimates, particularly for the shape parameter.*

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Figure 1: Comprehensive extreme value analysis of flood data: (a) Annual maximum flood time series showing block maxima with location parameter and 50-year return level; (b) GEV probability plot demonstrating goodness-of-fit to theoretical quantiles; (c) Fitted GEV density overlaid on empirical histogram of annual maxima with estimated shape parameter indicating moderate heavy tail; (d) Return level plot with 95% bootstrap confidence intervals showing extrapolation to 500-year event; (e) Daily flood observations with threshold exceedances highlighted for POT analysis; (f) Mean excess plot exhibiting approximate linearity above threshold, validating GPD approximation; (g) Threshold diagnostics showing number and rate of exceedances versus threshold value; (h) GPD fitted to excesses over threshold with estimated parameters; (i) Comparison of return levels from block maxima (GEV) and POT (GPD) approaches; (j) Shape parameter estimates from both methods compared to true value; (k) GPD probability plot for excess validation; (l) Tail probability comparison on logarithmic scale showing convergence of GEV and POT estimates in the far tail region.

Example 5.2 (Threshold Selection Diagnostics) *The mean excess plot (Figure **??**f) shows approximate linearity above $u =$?? m^3/s, validating the GPD model for excesses. Below this threshold, curvature indicates violation of the asymptotic approximation. The selected threshold yields ?? exceedances (??% of observations), balancing asymptotic validity against estimation precision.*

5.3 Applications

Example 5.3 (Flood Risk Assessment) *For infrastructure design with a 50-year design life and acceptable failure probability of 0.02, the required design flood is the* $(1 - 0.02)^{-1/50} \approx 122$*-year return level: ?? m^3/s (conservative estimate using 100-year level). This accounts for the risk of at least one exceedance during the structure's lifetime.*

6 Conclusions

This analysis demonstrates the application of extreme value theory to flood frequency analysis:

1. The Fisher-Tippett-Gnedenko theorem provides theoretical justification for the GEV distribution as the limiting distribution of block maxima, with fitted parameters $\mu =$ **??**, $\sigma =$ **??**, and $\xi =$ **??**
2. The 100-year return level is estimated as **??** m^3/s [**??**, **??**], indicating the flood magnitude expected to be exceeded on average once per century
3. The POT method with threshold $u =$ **??** m^3/s yields GPD parameters $\sigma_u =$ **??** and $\xi =$ **??**, with return levels agreeing within **??**% of the GEV estimates
4. The positive shape parameter indicates heavy-tail behavior, suggesting that extreme events significantly more severe than historically observed remain plausible
5. Mean excess plot diagnostics and probability plots confirm model adequacy for both GEV and GPD fits

Further Reading

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